Organization of Gene
“Mutations & Genome Variations”

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Molecular Biology :

Mutations:

- When it was recognized that changes (mutations) in genes occur spontaneously (T. H. Morgan, 1910) and can be induced by X-rays (H. J. Muller, 1927), the mutation theory of heredity became a cornerstone of early genetics.

- Genes were defined as mutable units, but the question what genes and mutations are remained.

- Today we know that mutations are changes in the structure of DNA and their functional consequences, and can be induced by **Biological chemical and Physical factors**.

- The study of mutations is important , where the mutations cause diseases, including all forms of cancer, as well without mutations, well-organized forms of life would not have evolved.

- Thus, they **“The Mutations”** represent a link between heredity and environment.

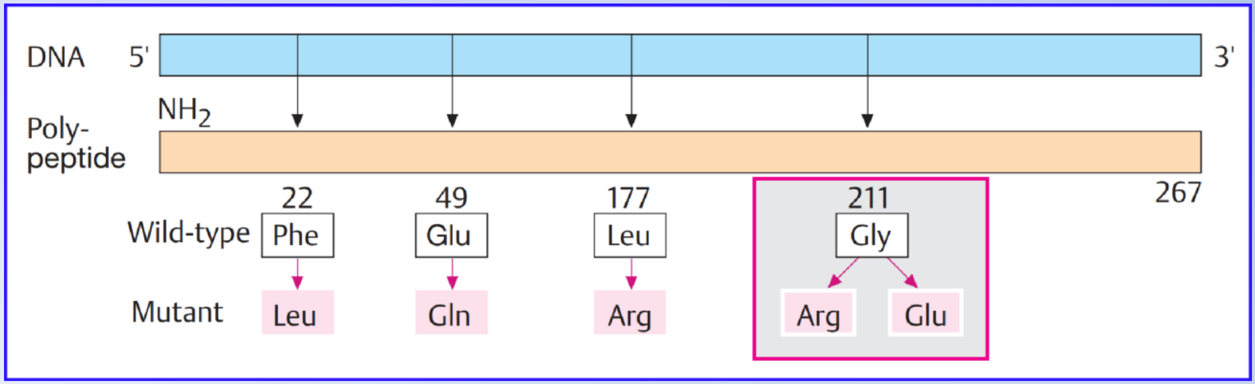
Molecular Biology :

Gene & Mutation:

- A mutation is a heritable change in the DNA sequence of an organism. The resulting organism, called a mutant, may have a recognizable change in phenotype compared to the wild type.

- An alteration (mutation) of the DNA base sequence may lead to a different codon and this will conferred to mRNA through transcription, and may lead to an altered amino acid sequence in a protein on translation.

- Because proteins carry out the majority of cellular functions, a change in amino acid sequence in a protein may lead to an altered phenotype for the cell and organism.

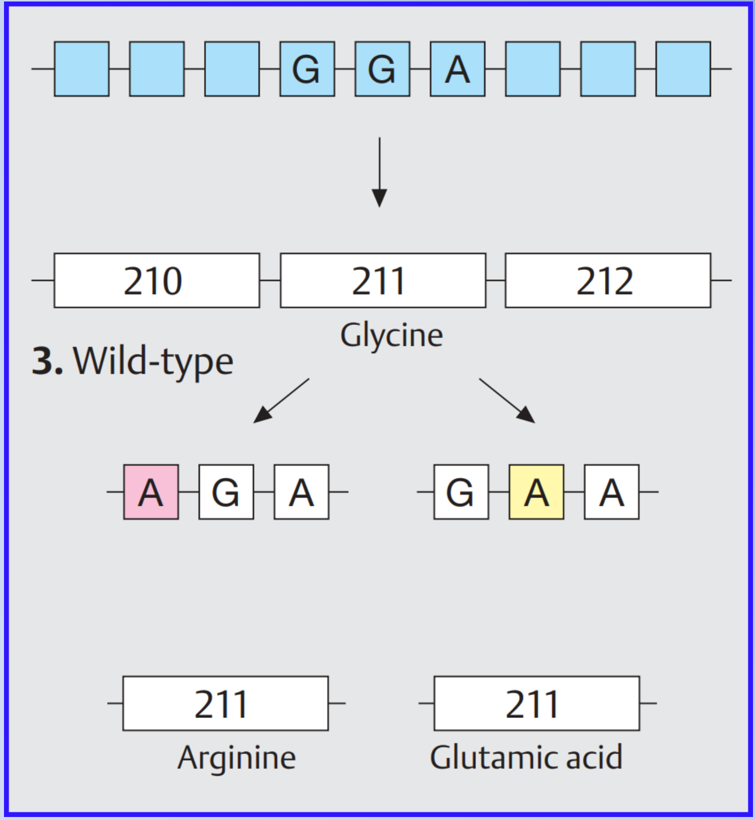


Gene & Mutation:

- Every mutation has a defined position. The position of the resulting change in the sequence of amino acids corresponds to the position of the mutation.

- Different mutations at one position (one codon) in different DNA molecules are possible.

- For example: Two different mutations have been observed In E. coli bacteria at position 211: glycine (Gly) to arginine (Arg) and Gly to glutamic acid (Glu).



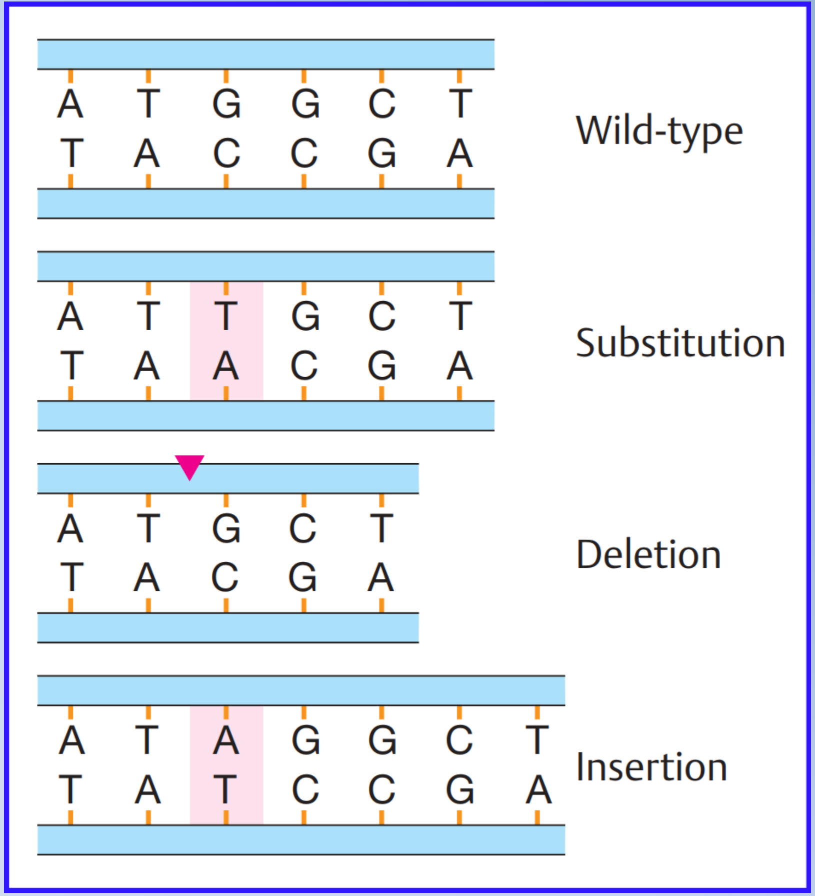
Normally (in the wildtype):

1. Codon 211 is GGA and codes for glycine.
2. A mutation of GGA to AGA leads to a codon for arginine.
3. A mutation to GAA leads to a codon for glutamic acid.

Types of Mutation:

Basically, there are three different types of mutation involving single nucleotides (point mutation):

1. Substitution (exchange),
2. Deletion (loss) “Frameshift mutation”
3. Insertion (addition) “Frameshift mutation”



With substitution the consequences depend on how a codon has been altered. Two types of substitution are distinguished:

- Transition (exchange of one purine for another purine or of one pyrimidine for another). ***C↔T or G ↔ A***
- Transversion (exchange of a purine for a pyrimidine, or vice versa). ***C↔A or G↔T***

Molecular Biology :

The chemical nature of mutations:
1. Error in replication

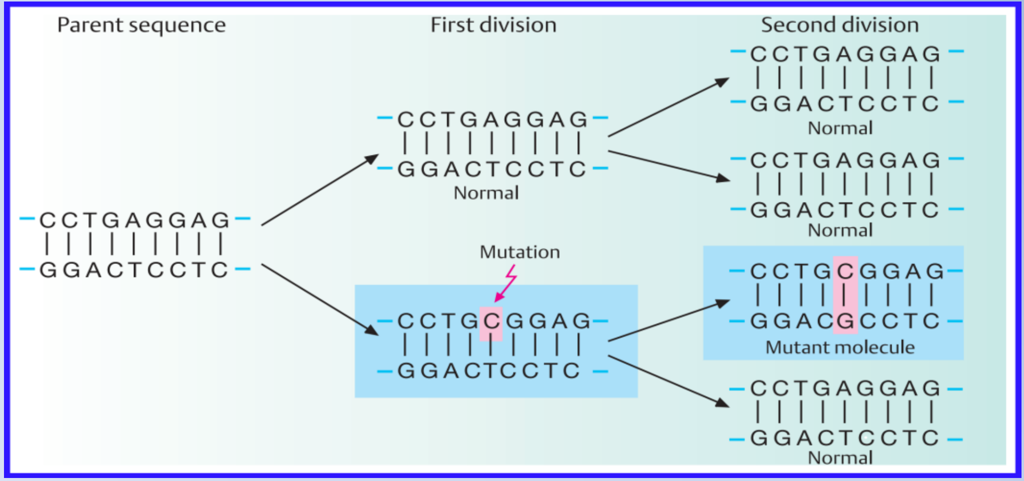
- The synthesis of a new strand of DNA occurs by semiconservative replication based on complementary base pairing.

- Errors in replication occur at a rate of about 1 in 10^5. This rate is reduced to about 1 in 10^7 to 10^9 by proofreading mechanisms.

- When an error in replication occurs before the next cell division (here referred to as the 1st division after the mutation), e.g., a (C) might be incorporated instead of an (A), the resulting mismatch will be recognized and eliminated by mismatch repair in most cases.

- However, if the error is undetected and allowed to stand, the next 2nd division will result in a mutant molecule containing a CG instead of an AT pair at this position.

- This mutation will be perpetuated in all daughter cells. Depending on its location within or outside of the coding region of a gene, functional consequences due to a change in a codon could result.



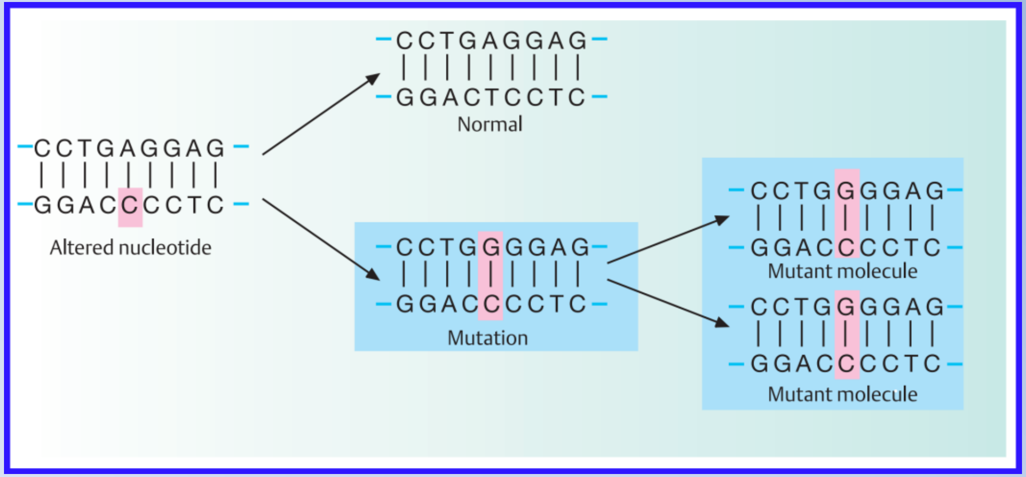
Molecular Biology :

The chemical nature of mutations:
2. Mutagenic alteration of a nucleotide

- A mutation may result when a structural change of a nucleotide affects its base-pairing capability.

- The altered nucleotide is usually present in one strand of the parent molecule.

- If this leads to incorporation of a wrong base, such as a C instead of a T, the next (second) round of replication will result in two mutant molecules.



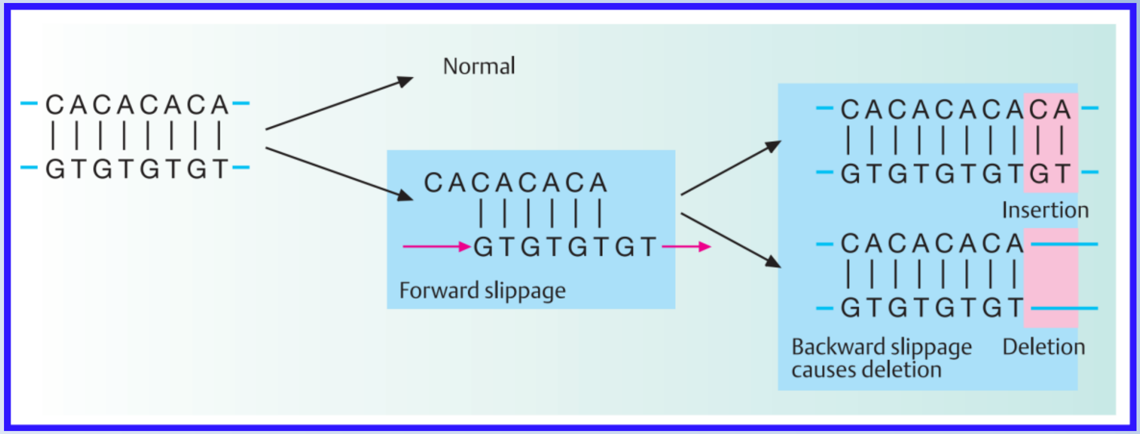
Molecular Biology :

The chemical nature of mutations:
3. Replication slippage

- A different class of mutations does not involve an alteration of individual nucleotides, but results from incorrect alignment between allelic or nonallelic DNA sequences during replication.

- When the template strand contains short repeats, e.g., CACACA, the newly replicated strand and the template strand may shift their positions relative to each other.

- The replication or polymerase slippage lead to incorrect pairing of repeats, some repeats are copied twice or not at all.



Molecular Biology :

The chemical nature of mutations:
2. Replication slippage “Show Video”

Molecular Biology :

DNA Repair

- The process of DNA replication is highly accurate, but mistakes can occur spontaneously or be induced by mutagens.
- Uncorrected mistakes can lead to serious consequences for the phenotype.
- Cells have developed several repair mechanisms to minimize the number of mutations that persist:

1. Proofreading:

- Most of the mistakes introduced during DNA replication are promptly corrected by most DNA polymerases through a function called proofreading.
- In proofreading, the DNA polymerase reads the newly added base, ensuring that it is complementary to the corresponding base in the template strand before adding the next one.
- If an incorrect base has been added, the enzyme makes a cut to release the wrong nucleotide and a new base is added

Molecular Biology :

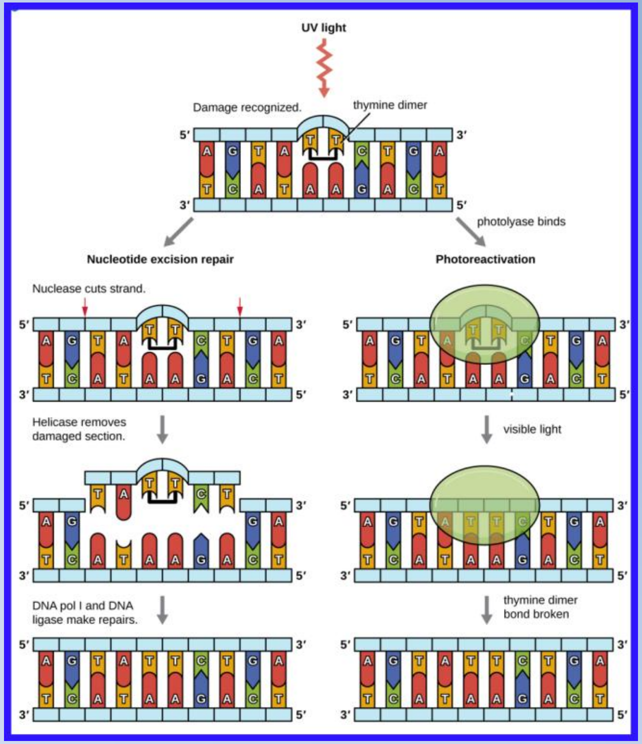
DNA Repair

2. Mismatch Repair:

- Some errors introduced during replication are corrected shortly after the replication machinery has moved. This mechanism is called mismatch repair.
- The enzymes involved in this mechanism recognize the incorrectly added nucleotide, remove it, and replace it with the correct base.

3. Repair of Thymine Dimers:

- Because the production of thymine dimers is common (many organisms cannot avoid ultraviolet light).
- Mechanisms have evolved to repair these lesions:
  a. Light repair: Photolyase / Require a Visible Light
  b. Dark repair: In nucleotide excision repair, enzymes remove the pyrimidine dimer and replace it.



Genetic Variation:

The gene(s) directly responsible for the observed characteristic may be absolutely identical but the effects may be different because of variation in other genes that affect their expression, or because of alteration in other cellular components that affect the activity of the proteins encoded by those genes

The study of genetic variation can be used to examine differences between members of the same species, such as:

- The study of bacterial characteristics (such as antibiotic resistance)
- Investigation of human genetic diseases.
- Differentiate between individuals (for example in forensic analysis).
- Compare the genetic composition of members of different species, which can throw invaluable light on the processes of evolution.
- Helping to define the taxonomic relationship between species.

Molecular Biology :

Single nucleotide polymorphisms (SNPs):

- All individual humans share genome sequences that are approximately 99.9% the same. The remaining variable 0.1% is responsible for the genetic diversity between individuals.

- Genetic polymorphism is the existence of variants with respect to a gene locus (alleles) .

- DNA polymorphisms are important as genetic markers to identify and distinguish alleles at a gene locus and to determine their parental origin.

- Single nucleotide polymorphisms (SNPs) (pronounced “snips”), where two or more possible nucleotides occur at a specific mapped location in the genome, are the most common type of genomic sequence variation among individuals.

- These allelic variants differ in a single nucleotide at a specific position. At least one in a thousand DNA bases differs among individuals